

- **Problem Statement ID** – [Enter PS ID]
- **Problem Statement Title** – Automated Road Obstruction & Hazard Mapping from Street/Dashcam Video
- **Team Name** – Team UniVerse
- **Team Leader Name** – [Enter team leader name]
- **Members Names** – [Enter member names]

Our solution

TerraTrace

IDEA TITLE: TerraTrace

Proposed solution: dashcams become hazard sensors; every hazard becomes a tracked case.



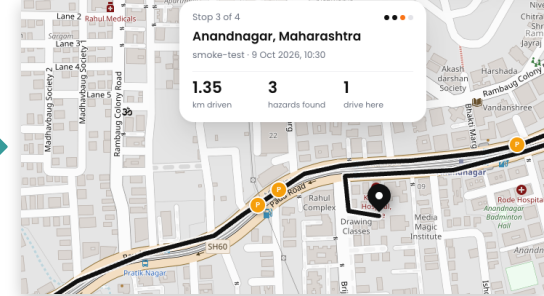
1 Camera sees the road

Helmet/dash camera on NH3, Himachal



2 AI spots the hazard

Water across the road: flooding, 96%



3 Mapped & tracked

Route, exact pins, case for the department



Sees what satellites miss

Floods, landslides, fallen trees, potholes under canopies & overpasses



Exact hazard location

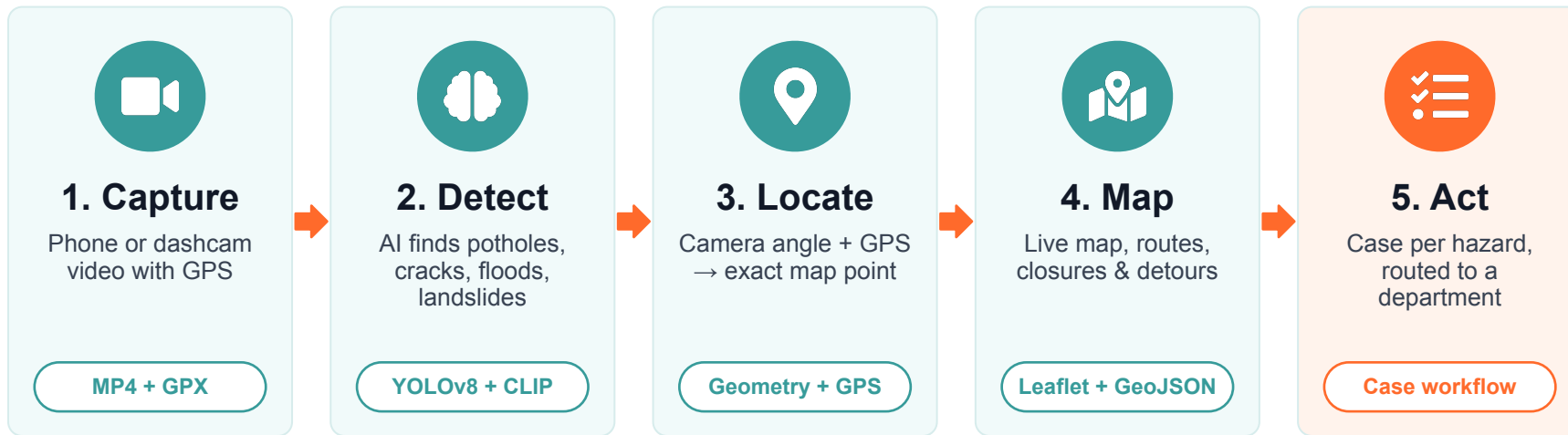
Camera geometry pins the hazard itself, not just the car



Detection → resolution

Each hazard is a case: priority, department, deadline, status

How it works, in five steps



Built with

Python

YOLOv8

YOLO-World

CLIP

OpenCV

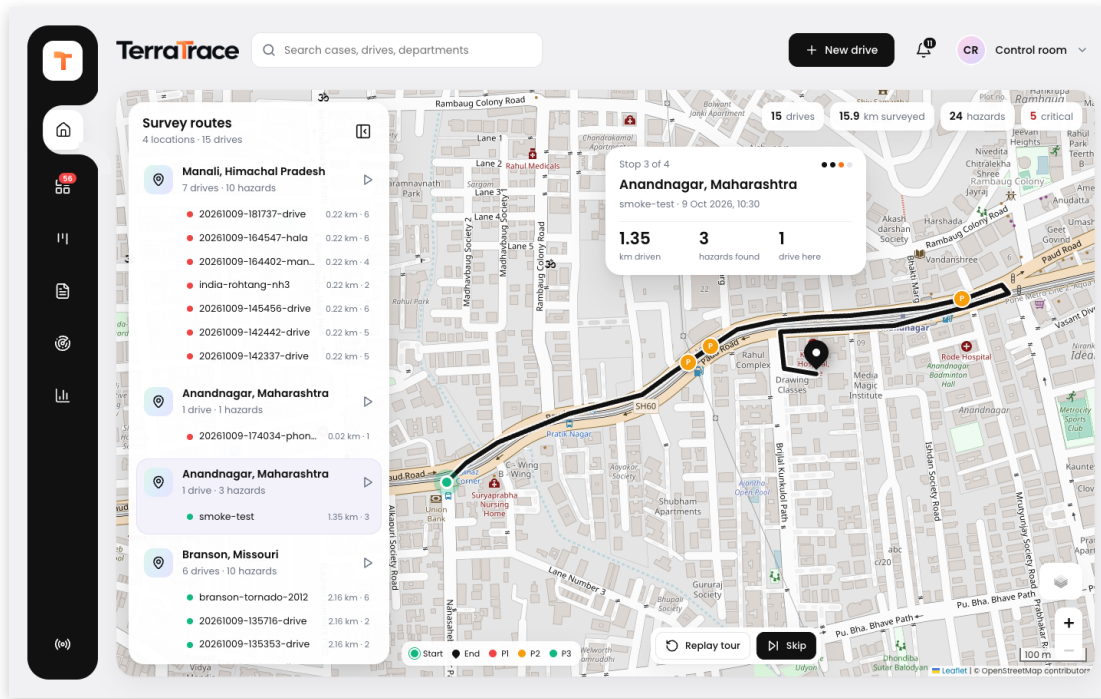
FastAPI

SQLite

Leaflet

GeoJSON

 **Runs on a laptop GPU (Apple M5): no cloud needed.** Three ways in: upload a drive, replay it live, or stream a phone camera.



TerraTrace Home: an animated tour draws every analysed route from start to end and drops hazard pins as it passes them.

22 fps AI speed on a laptop GPU

4× faster than real time

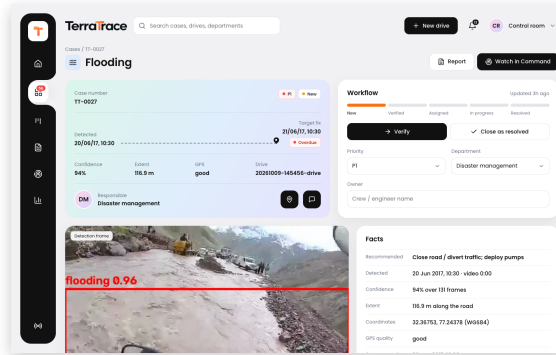
3 inputs: upload · live replay · phone

16/16 test checks on 3 real clips



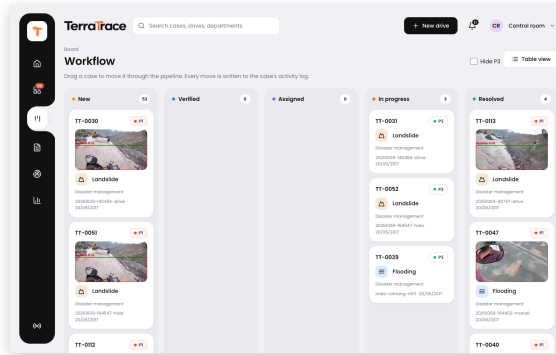
Evidence photo saved with every hazard

From detection to resolution: one platform for the control room



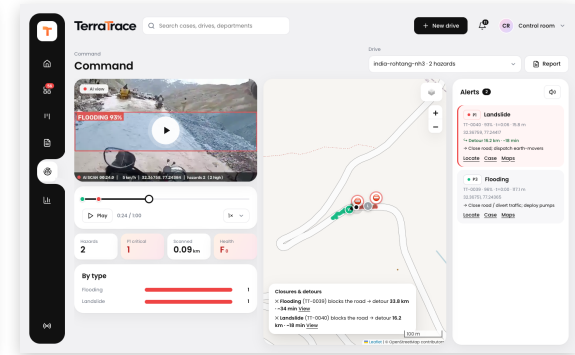
Case tracking

Detected → target fix, priority, department, evidence, notes



Workflow board

New → Verified → Assigned → In progress → Resolved



Command

AI-view video synced with the map; live alerts & detours



Also: one-click incident report (PDF) for authorities · road health grade per drive · GeoJSON / CSV export for any GIS · phone-as-dashcam live mode

Feasibility analysis



Already working end-to-end

Upload, live replay & phone streaming → mapped cases with a full workflow



No special hardware

Runs on a laptop GPU; any phone or dashcam is the sensor



Open data & models

Pretrained on RDD2022 (47k+ images); open-source stack, no licence cost

Challenges & risks → our strategies

Challenge / risk	Strategy
AI misses or invents hazards	Whole-road CLIP scene check finds floods & landslides; dry roads and hillsides are filtered out; fine-tune next
Intermittent / noisy GPS	Time interpolation + heading smoothing; GPS-gap flag on every hazard
Unknown camera mounting	Camera presets (phone / dashcam / helmet) + automatic tilt from the horizon
Same hazard seen many times	Object tracking + spatial clustering → one case per real hazard
Phone can't reach the laptop	Single HTTPS server; phone hotspot fallback



Municipal corporations

Pothole & road-damage inventory from buses, garbage trucks, taxis; repairs prioritised and tracked to closure



Disaster response teams

Within hours of a storm or landslide: which roads are blocked or flooded, and the detour around them



Citizens & commuters

Safer routing, fewer accidents and vehicle damage; transparent public hazard map

Key benefits

- **Low cost** — reuses phones & dashcams already on the road
- **Fast** — hazard map in minutes, not days of surveys
- **Accountable** — every hazard has an owner, a deadline and a status
- **Interoperable** — standard GeoJSON for any GIS or city dashboard



Tested on an Indian highway (NH3, Rohtang): flooding detected, road closed, detour found

1

RDD2022 road damage dataset

[Arya et al., arXiv:2209.08538](#) · github.com/sekilab/RoadDamageDetector

2

comma2k19 driving dataset (video + GNSS)

github.com/commaai/comma2k19

3

YOLOv8 / YOLO-World (Ultralytics)

docs.ultralytics.com · [Cheng et al., CVPR 2024](#)

4

RDD-trained YOLOv8s weights

github.com/oracl4/RoadDamageDetection

5

CLIP: zero-shot image–text matching

[Radford et al., ICML 2021](#) · github.com/openai/CLIP

6

ByteTrack multi-object tracking

[Zhang et al., ECCV 2022](#)

7

DBSCAN density clustering

Ester et al., KDD 1996

8

GeoJSON · Leaflet · OpenStreetMap

[IETF RFC 7946](#) · leafletjs.com · openstreetmap.org

9

OSRM routing · Nominatim geocoding

project-osrm.org · nominatim.org (detours, place names)

10

TerraTrace source code

github.com/aryan270506/Team-UniVerse-Geo-AI

11

Indian road footage (NH3 demo)

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